



## 8 MECHANICAL SYSTEMS

It is of great importance that the operator knows the equipment as a whole and understands the operation of each set. This knowledge will bring several benefits such as easier operation and adjustment of the machine, easy to perform small maintenance, find and correct possible defects, order parts for replacement or maintenance, among others.



**Caution: When performing setup and adjustment changeover procedures, keep a distance from components that have moving parts.**

### 8.1 Main Components And Their Functions

➤ **Film Passage:**

Device for passing the film, which directs the film to the forming tube.

➤ **Vertical Seal:**

Device responsible for "welding" / "sealing" the bale in the vertical direction.

➤ **Vertical Cooling:**

It is responsible for cooling the package in the vertical weld region after sealing, by means of an air jet.

➤ **Film Drive Belt**

Device that drags the film from the forming tube.

➤ **Jaw:**

Device responsible for "clamping" the package to perform horizontal welding and apply cooling. The horizontal, back and front welding metals, knife, and horizontal cooling channels are housed in the clamp.

➤ **Horizontal Seal:**

Device that has the function of "sealing" the package in the horizontal direction, both in the bottom and in the top (bottom and top of the bale). The cutting blade is fixed to the horizontal welding set, that is, the horizontal welding and the cutting are performed simultaneously, dividing the bales (top and bottom).

➤ **Horizontal Cooling:**

Responsible for cooling the package in the horizontal seal after the weld region by means of the air jet.

➤ **Forming Tube:**

Component that forms the package, that is, shapes the film and defines its width and depth.

➤ **Package Expeller:**

Responsible for the removal of the package, propelling it up the collecting conveyor.

➤ **Motorized Centralizer:**

The Motorized Centralizer through a motor is responsible for centralizing the plastic coil in the machine, when it tends to decentralize.

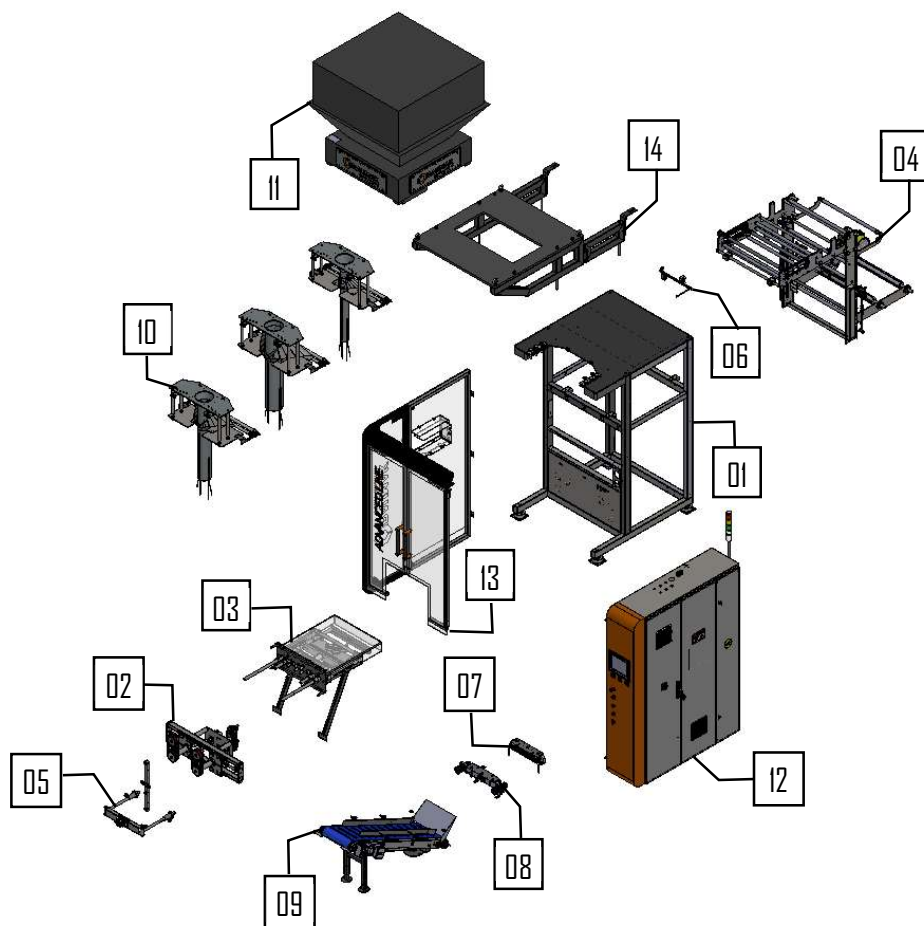


## 8.2 Identification Of Principal Components

To facilitate the identification of the sets that make up the MULTI BAG G, below is an illustrative image with the main sets identified.

On the following pages you will find important details and information on the operation, assembly and maintenance of the individual assemblies.

The images below are only illustrative, check the model and components acquired in the purchase of the equipment.



1. Structure;
2. Film Drive Belt;
3. Jaw;
4. Unroll Spool;
5. Vertical Seal;
6. Centralizer;
7. Horizontal Seal Ruler;

8. Laminate Horizontal Seal;
9. Out Put Conveyor;
10. Forming Tube;
11. Weigher;
12. Electrical Panel;
13. Doors and Guards;
14. Over Ceiling Weigher Base.



### 8.2.1 Jaw

In the open-closed cabinet the actuation is done by a cylinder, through a pneumatic valve. The enabling is done by the PLC. It programs the times and counters that will turn on the electro-pneumatic valve that will drive this cylinder.

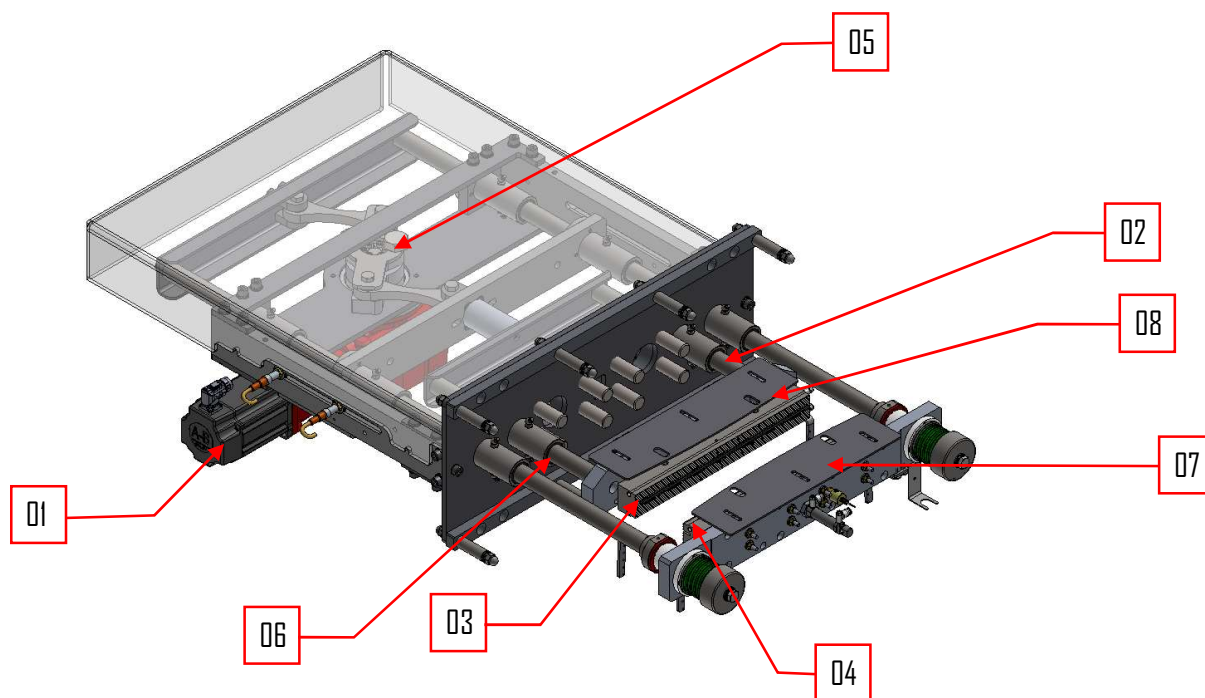
With the plastic held between the jaws stretched and firm, the package is welded and cut, controlled by the PLC through an electropneumatic valve that activates the pneumatic cylinder. The PLC programs the activation time and, through the panel temperature controllers, the welding power. Note that the sealing of the top and bottom (of the next package) is done simultaneously with the cut.

When the electro-pneumatically retracting the jaws, the horizontal cooling is activated, through the PLC program, being activated for a certain time. The objective is to make the weld cool quickly, preventing it from wrinkling or opening.

For a good functioning of the welding system with tubular resistance, it is necessary to observe some cares:

- Keep the jaw grooves always clean;
- Keep the jaws aligned; possible misalignment may be caused by the wear of the sliding bushings of the open and close set.

Obs: Horizontal cooling is an optional item.



1. Servo Motor;
2. Metal and knife drive cylinder;
3. Rear Welding Metal;
4. Front Weld Metal;
5. Gag drive rods;
6. Slide bearing;
7. Front jaw;
8. Rear jaw.



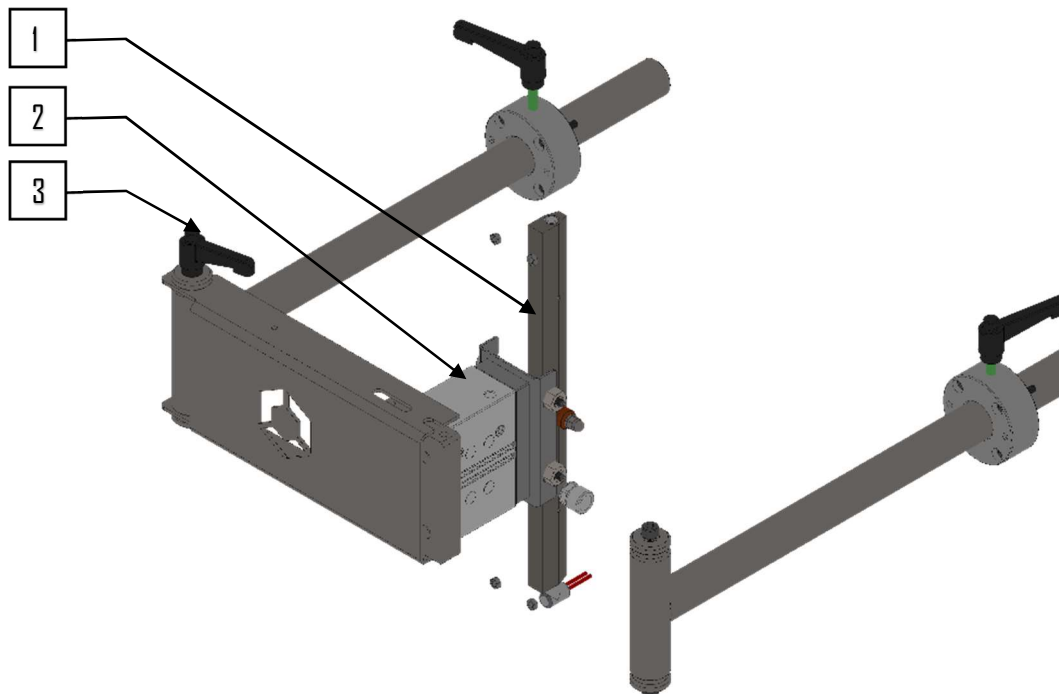
**Danger: Never perform any type of manual procedure near the clamps while the equipment is pressurized. Whenever manual intervention is required, end the cycle, press emergency and then you can perform the procedure safely.**

### 8.2.2 Vertical seal

The vertical welding is responsible for the longitudinal closing of the package formed by the plastic film on the tube. The movement of the hot bar weld (1) is electro-pneumatic, the PLC is responsible for activating the valve that actuates this cylinder (2). It is necessary to parameterize in the PLC the time in which the resistance will be actuated, and also how long it will remain actuated.

The vertical weld has a resistor (3), fed by a low voltage and high current transformer. This transformer supplies voltage and current to the Vertical weld. The temperature is controlled by the temperature controller on the panel.

A Teflon film is glued on the forming tube, serving as a support and also to mold the weld.



1. Vertical seal;
2. Actuator cylinder;
3. Opening latch.



**Danger: Never perform any type of manual procedure near the clamps while the equipment is pressurized. Whenever manual intervention is required, end the cycle, press emergency and then you can perform the procedure safely.**

## 8.3 Adjustmens and Format Changes

This section of the manual is dedicated to helping operators with the mechanical adjustments that can be made during the operation of the machine, improving its performance. The adjustments described below, which will be carried out through the gauntlets and regulators, which are not considered maintenance and can be carried out by the operator, provided they have the knowledge, experience and it is completely safe for perform this action. All the operations described below if applied incorrectly will cause the machine to malfunction.



**Danger: Never do any type of manual procedure with the pressurized equipment. Whenever manual intervention is necessary, end the cycle, emergency press and then the procedure can be performed safely.**

### 8.3.1 Pneumatic adjustments

#### 8.3.1.1 FLOW REGULATORS IN ACTUATORS

All pneumatic actuators have air flow regulators in their forward and reverse. These have the function to adjust the actuator stem speed, allowing a smooth operation. This regulation can be performed through the handles found in the inputs and outputs of pneumatic actuators.

#### 8.3.1.2 PRESSURE REGULATOR

The machine is equipped with pressure regulators in its pneumatic system. These are applied specifically to the main air inlet and to assemblies that require precision and speed in the actuators. Through a handle on the regulator body, the operator can increase and decrease the system pressure as needed.

### 8.3.2 Temperature control on seals

To achieve good product quality, it is necessary that the machine performs resistant and good-looking welds. For this, the machine has a welding temperature control system, but this function is only accessed through the temperature controllers on the equipment's panel.

It is very important that the operator knows these parameters, because periodically the package welds should be checked and, if necessary, the temperature adjusted.



## 8.4 Basic Operating Sequence

With the equipment previously installed, proceed according to the following start-up instructions:

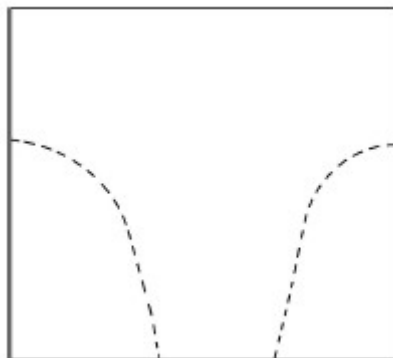
1. With the electrical and pneumatic installation already completed by the authorized Raumak technician, perform the film reel installation procedure;
2. Check that there are no loose objects on the machine;
3. Turn on the main switch;
4. Check air pressure on the machine's road (6 - 7 bar);
5. Close all equipment doors;
6. Press the "START" button. Warning: The machine will preheat and then drive the welding metals, horizontal welding (or knife).
7. Press the "STOP" button to stop the machine within the cycle.

**WARNING: Any abnormality in operation press the emergency button.**

Press the EMERGENCY button in case of any problems during operation. In case of any anomaly that cannot be resolved by reading this manual, please contact us at 55 47 3370-4540 and speak to one of our technicians.

### 8.4.1 Film core replacement procedure

1. Initially shut down the machine cycle and operate the emergency button so that there is no danger;
2. Check if the reel to be changed is finished. In case it's not finished, rewind the stretch that is inside the machine or make a cut next to the film reel, leaving it free to be removed;
3. Access the rear part of machine, raise the unwinder (bambo roll) to have access to the removal of the bobbin shaft;
4. With the bobbin shaft free, in two people, taking one at each end, remove the shaft and bobbin kit and deposit them on some safe base or even on the ground.
5. Loosen and remove the left nut;
6. From the right side, remove the shaft from inside the coil;
7. Doing the opposite procedure, insert the shaft into the filled reel from the right side. Attention, the position and the unrolling direction of the film can't be changed, that is, observe the old reel before disassembling it, or follow the illustration of topic 15;
8. Put the left nut back on and tighten it;
9. Again, in two people, insert the shaft into the machine's supports carefully. Note that the shaft seats should fit over the bearings on both sides, allowing smooth rotation;
10. With the aid of some measuring instrument, check if the coil is centered. If necessary, center it through the right and left nuts;
11. With a cutting blade, make an angular cut forming a point at the end of the film, as in the illustration below, in order to facilitate the passage of the plastic film through the collar.

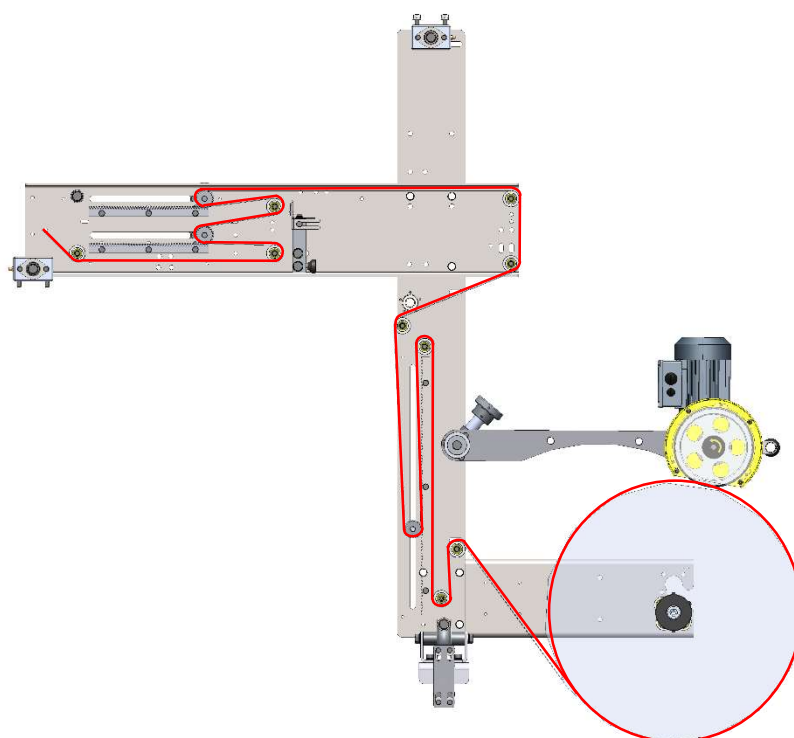


12. Based on the illustration of the topic 15, pass the film through the rollers following the correct path to the forming tube. Check if the reel is unwinding in the correct position and direction, comparing the scheme of topic 15 and the direction stickers of the rollers;
13. Place the film on the collar passing the tip through the front of the collar. Manually pull a quantity of film so that it settles. Adjust the film manually until it is forming the perfect package;
14. To start the machine production cycle, follow the procedure in the item 8.4 of this manual (Basic Sequence of Operation).





15. Film passing scheme for MULTI BAG G:



## 8.5 Operation

### 8.5.1 Panel and command

Aiming to meet customer expectations and ensure agility and convenience in operation, this equipment carries a technology of Interaction between Man and Machine, the HMI. This HMI, communicating with the PLC, is capable of controlling all machine parameters. Therefore, the equipment's general commands are made through a central operation panel, where the start, stop, reset, cycle stop, emergency and HMI operation interface buttons are located on it.

Through the HMI, the operator will be able to control and adjust the machine's functions, change recipes, enable functions, check error messages and even monitor production statistics. This control is presented in several screens that will be explained below.

### 8.5.2 Start button

The Start button is responsible for starting the machine cycle.

### 8.5.3 Stop button

The Stop button is responsible for closing the current cycle instantly.

### 8.5.4 Reset button

The reset button is responsible for rehabilitating the machine after an Emergency stop or silencing some specific alarms.

### 8.5.5 Emergency button

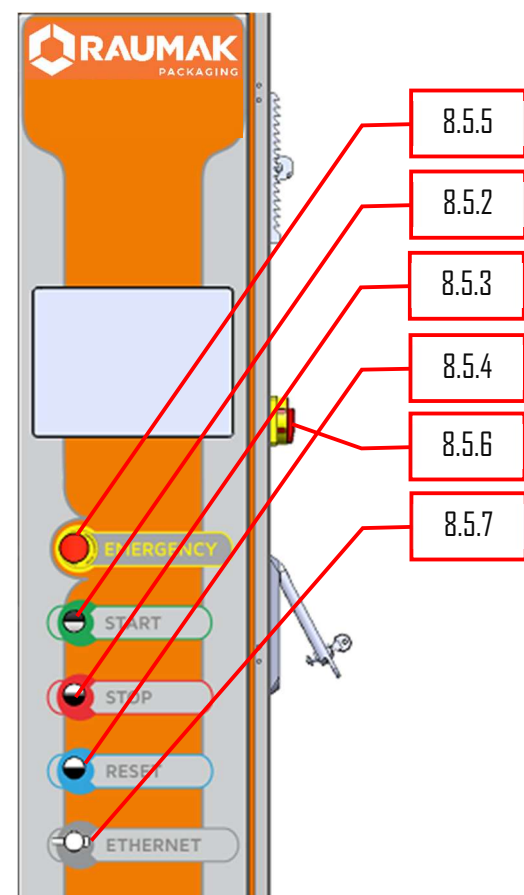
Emergency button is responsible for stopping the machine instantly and depressurizing it. This button should be pressed only in case of emergency.

### 8.5.6 General key

The general key (General Disconnect) allows you to activate and deactivate the supply of electrical energy for the machine.

### 8.5.7 Ethernet connection

Plug for an ethernet network cable connection





## 8.6 Alarm List

| Message                                                             |
|---------------------------------------------------------------------|
| Unroll inverter failure                                             |
| Fault jaw servo /*S:0 {[CLP]Stl_falha_servo_mordaca}*/              |
| Fault servo pulling belts/*S:0 {[CLP]Stl_falha_servo_tracionador}*/ |
| Inverter failure/*S:0 {[CLP]STL_Descricao_falha_inv_1}*/            |
| Panel emergency button active                                       |
| Front doors open                                                    |
| Side door open                                                      |
| Locked coil                                                         |
| Dater failure                                                       |
| Low air pressure                                                    |
| Eye mark failure                                                    |
| Contactor thermal disarmed                                          |
| Off-temperature horizontal welding                                  |
| Safety relay fault                                                  |
| Off-temperature vertical welding                                    |
| Loose coil                                                          |
| SERV1 Without power supply                                          |
| SERV2 Without power supply                                          |
| Film end                                                            |